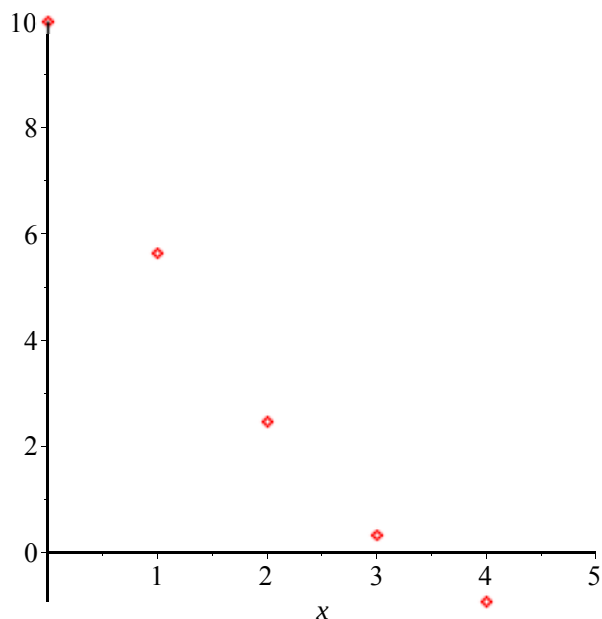


8_ Interpolation
Variant-2
2A-29 Bogdanov D.O.
Function 1

1) Perform linear, quadratic and cubic interpolation using the Lagrange polynomial.

a) Linear according to Lagrange:

```
>  
> restart;  
> with(plots):  
> X:=[0,1,2,3,4];  
X:= [0, 1, 2, 3, 4] (1)  
> Y:=[10,5.641667,2.466667,0.3250000,-0.933330];  
Y:= [10, 5.641667, 2.466667, 0.3250000, -0.933330] (2)  
> q1:=plot([X[i],Y[i]]$i=1..5,x=0..5,style=point,color=red):  
> display(q1);
```



```
> x0:=X[3];x1:=X[4];y0:=Y[3];y1:=Y[4];
```

x0:=2

x1:=3

y0:=2.466667

y1:=0.3250000

(3)

```
> L:=y0*(x-x1)/(x0-x1)+y1*(x-x0)/(x1-x0);
```

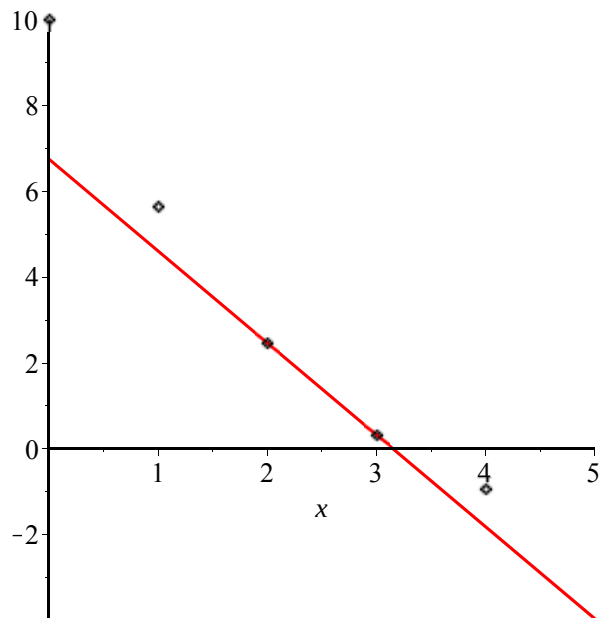
L:= -2.1416670 x + 6.7500010

(4)

```
> q1:=plot(L,x=0..5,color=red):
```

```
> q2:=plot([X[i],Y[i]]$i=1..5,x=0..5,style=point,color=black):
```

```
> display(q1,q2);
```



```
> x:=2.3;L;
```

```
x := 2.3
1.82416690
```

(5)

```
> x:=2.7;L;
```

```
x := 2.7
0.96750010
```

(6)

```
> x:='x';
```

```
x := x
```

(7)

b) Quadratic according to Lagrange:

```
> restart;
```

```
> with(plots):
```

```
> X:=[0,1,2,3,4];
```

```
X:= [0, 1, 2, 3, 4]
```

(8)

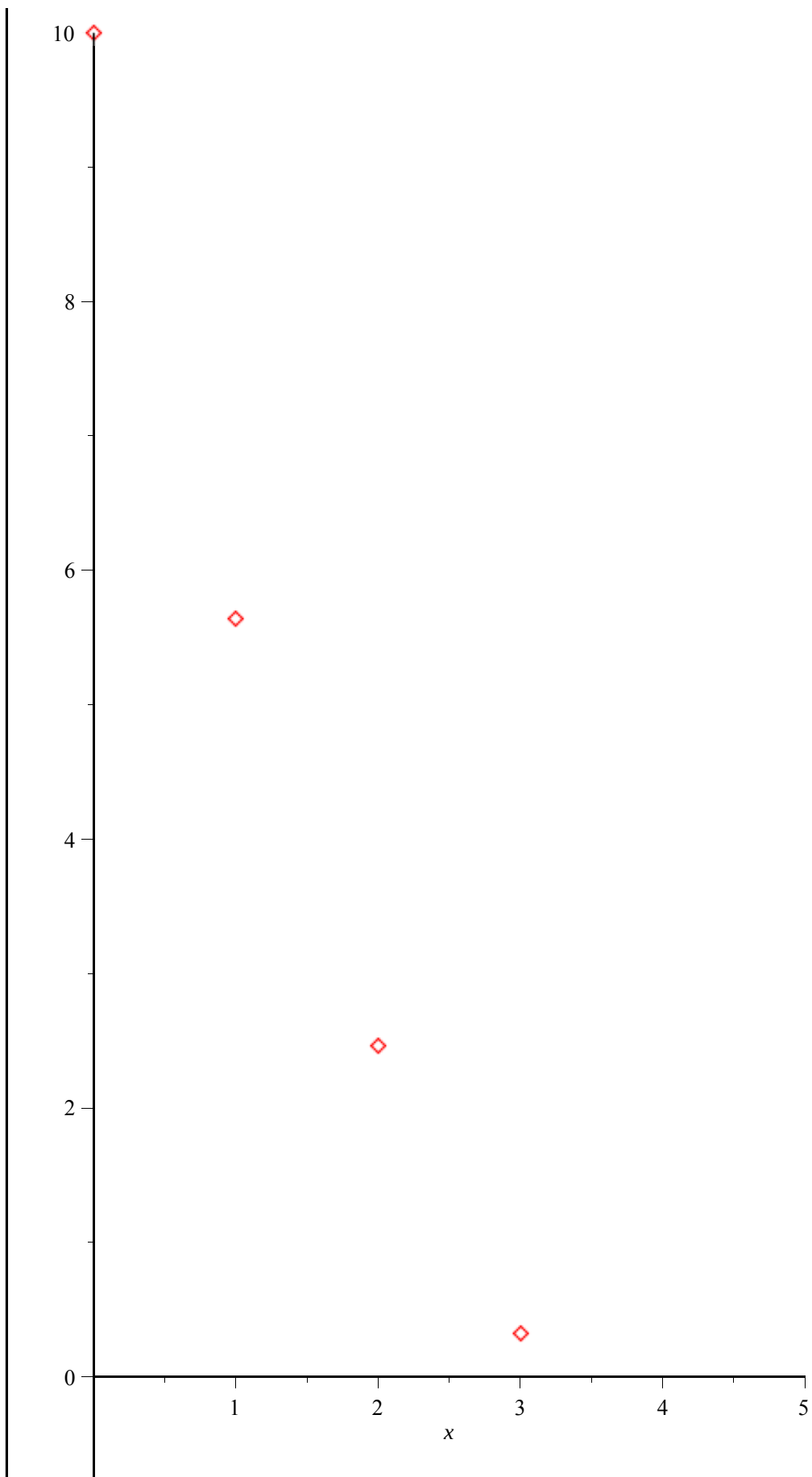
```
> Y:=[10,5.641667,2.466667,0.3250000,-0.933330];
```

```
Y:= [10, 5.641667, 2.466667, 0.3250000, -0.933330]
```

(9)

```
> q1:=plot([X[i],Y[i]]$i=1..5,x=0..5,style=point,color=red):
```

```
> display(q1);
```



```

> x0:=X[2];x1:=X[3];x2:=X[4];y0:=Y[2];y1:=Y[3];y2:=Y[4];
      x0:=1
      x1:=2
      x2:=3
      y0:=5.641667
      y1:=2.466667
      y2:=0.3250000
(10)

```

```

> L:=y0*((x-x1)*(x-x2))/((x0-x1)*(x0-x2))+y1*((x-x0)*(x-x2))/((x1-x0)*(x1-x2))+y2*((x-x0)*(x-x1))/((x2-x0)*(x2-x1));
L:=2.820833500 (x-2) (x-3) - 2.466667 (x-1) (x-3) + 0.1625000000 (x-1) (x-2)
(11)

```

```

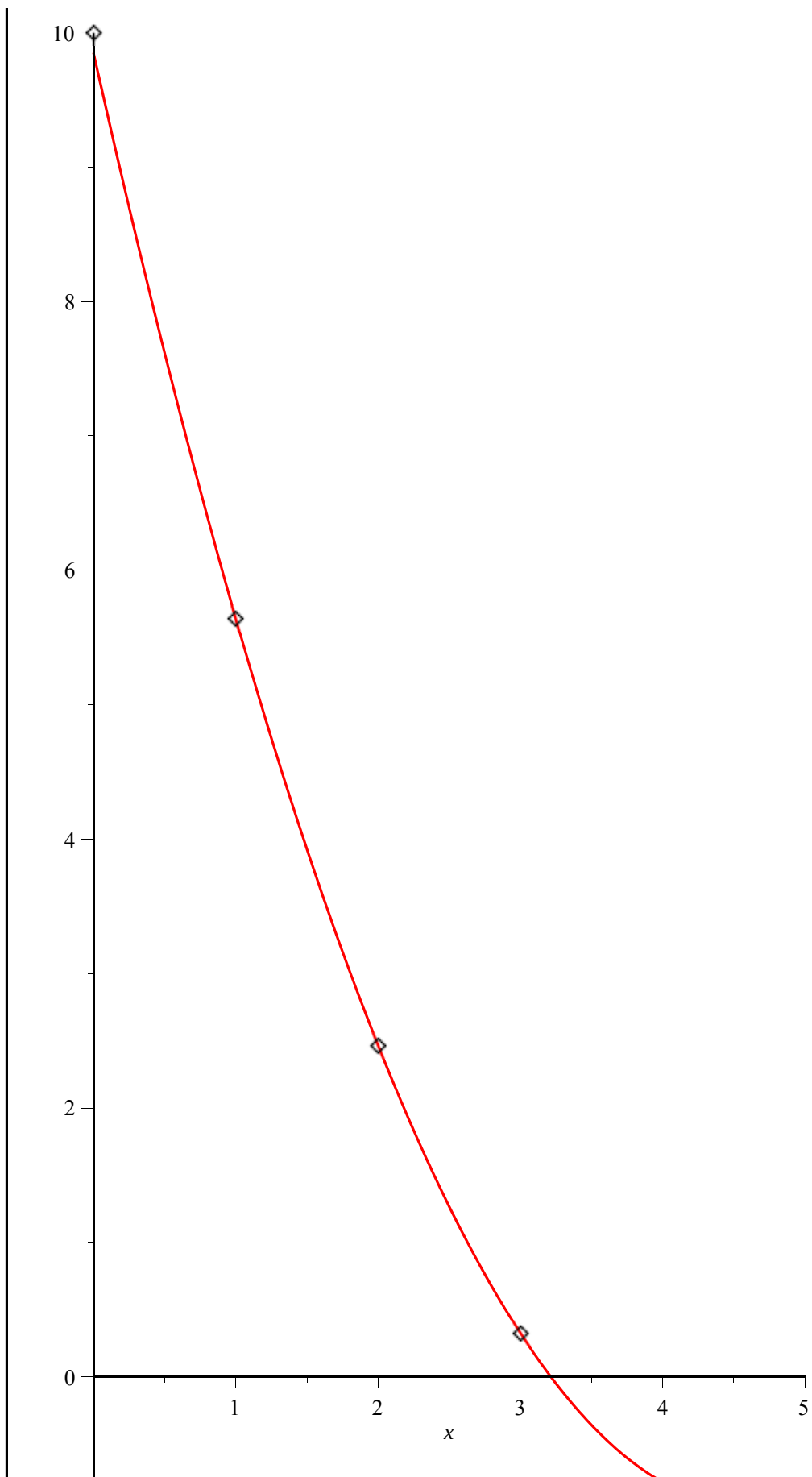
> simplify(L);
      0.516666500 x2 - 4.72499950 x + 9.850000000
(12)

```

```

> q1:=plot(L,x=0..5,color=red):
> q2:=plot([[X[i],Y[i]]$i=1..5],x=0..5,style=point,color=black):
> display(q1,q2);

```



```
> x:=2.3;L;
x := 2.3
1.715666935 (13)
```

```
> x:=2.7;L;
x := 2.7
0.8590001350 (14)
```

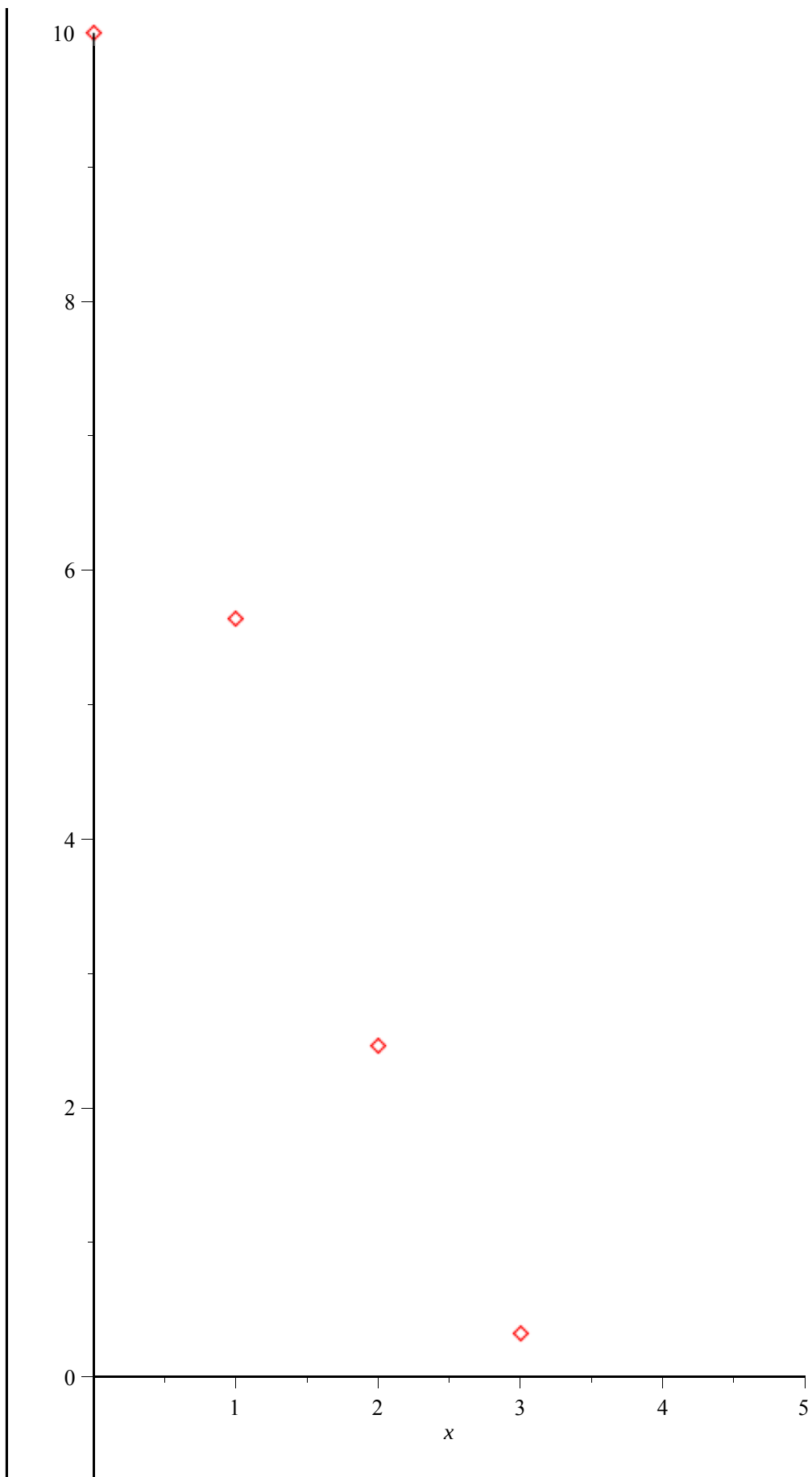
```
> x:='x';
x := x (15)
```

c) Cubic according to Lagrange:

```
> restart;
> with(plots):
> X:=[0,1,2,3,4];
X := [0, 1, 2, 3, 4] (16)
```

```
> Y:=[10,5.641667,2.466667,0.3250000,-0.933330];
Y := [10, 5.641667, 2.466667, 0.3250000, -0.933330] (17)
```

```
> q1:=plot([X[i],Y[i]]$i=1..5,x=0..5,style=point,color=red):
> display(q1);
```



```
> x0:=X[2];x1:=X[3];x2:=X[4];x3:=X[5];y0:=Y[2];y1:=Y[3];y2:=Y[4];
y3:=Y[5];
```

```
x0:=1
x1:=2
x2:=3
x3:=4
y0:=5.641667
y1:=2.466667
y2:=0.3250000
y3:=-0.933330
```

(18)

```
> L:=y0*((x-x1)*(x-x2)*(x-x3))/((x0-x1)*(x0-x2)*(x0-x3))+y1*((x-
x0)*(x-x2)*(x-x3))/((x1-x0)*(x1-x2)*(x1-x3))+y2*((x-x0)*(x-x1)*
(x-x3))/((x2-x0)*(x2-x1)*(x2-x3))+y3*((x-x0)*(x-x1)*(x-x2))/((
(x3-x0)*(x3-x1)*(x3-x2)));
```

```
L:=-0.9402778333 (x-2) (x-3) (x-4) +1.2333333500 (x-1) (x-3) (x-4)
-0.1625000000 (x-1) (x-2) (x-4) -0.1555550000 (x-1) (x-2) (x-3)
```

(19)

```
> simplify(L);
```

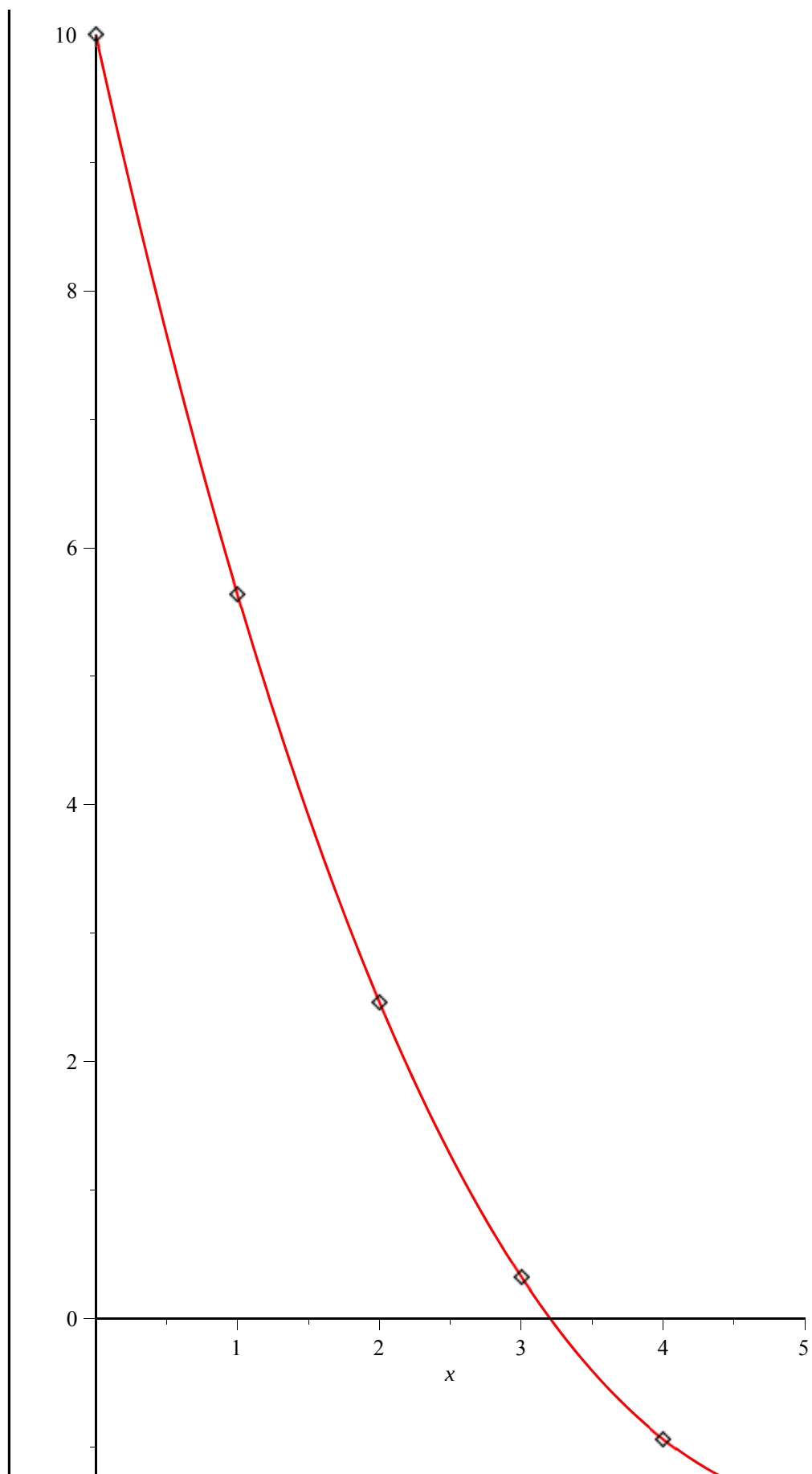
```
-0.024999333 x3 +0.666662500 x2 -4.99999217 x +9.999996000
```

(20)

```
> q1:=plot(L,x=0..5,color=red):
```

```
> q2:=plot([X[i],Y[i]]$i=1..5,x=0..5,style=point,color=black):
```

```
> display(q1,q2);
```

```
> x:=2.3;L;
```

$$x := 2.3$$

$$1.722491753$$

(21)

```
> x:=2.7;L;
```

$$x := 2.7$$

$$0.8679248970$$

(22)

```
> x:='x';
```

$$x := x$$

(23)

```
> x:=2.7;N;
```

$$x := 2.7$$

$$0.8679249744$$

(24)

```
> x:='x';
```

$$x := x$$

(25)

```
> `?`
```

$$`?`$$

(26)

2) Perform interpolation using the spline command

a) Linear:

```
> restart;
```

```
> with(plots) :
```

```
> X := vector([0, 1, 2, 3, 4]);
```

$$X := \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix}$$

(27)

```
> Y := vector([10, 5.641667, 2.466667, 0.3250000, -0.9333330]);
```

$$Y := \begin{bmatrix} 10 & 5.641667 & 2.466667 & 0.3250000 & -0.9333330 \end{bmatrix}$$

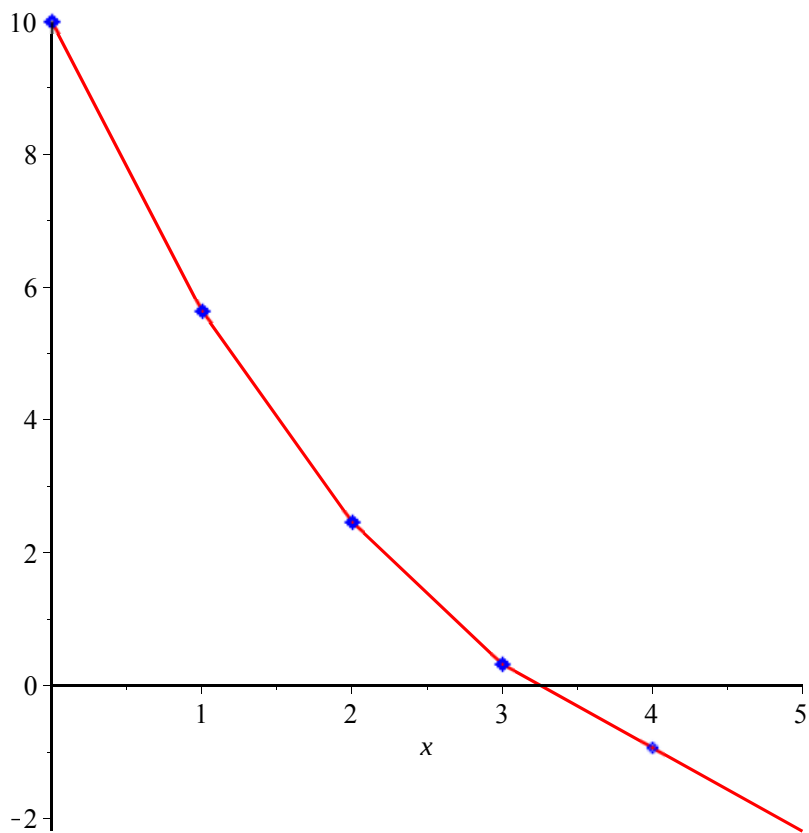
(28)

```
> f[1] := spline(X, Y, x, linear);
```

$$f_1 := \begin{cases} 10 - 4.358333 x & x < 1 \\ 8.816667 - 3.175000 x & x < 2 \\ 6.7500010 - 2.1416670 x & x < 3 \\ 4.09999900 - 1.2583300 x & \text{otherwise} \end{cases}$$

(29)

```
> plot([ f[1], [X[i], Y[i], i = 1 ..5]], x=0 ..5, style=[line, point], color=[red, blue]);
```



```
> x:=2.3;f[1];
```

```
x := 2.3
1.82416690
```

(30)

```
> x:=2.7;f[1];
```

```
x := 2.7
0.96750010
```

(31)

```
> x:='x';
```

b) Quadratic:

```
x := x
```

(32)

```
> restart;
```

```
> with(plots):
```

```
> X := vector([0, 1, 2, 3, 4]);
```

```
X := [ 0 1 2 3 4 ]
```

(33)

```
> Y := vector([10, 5.641667, 2.466667, 0.3250000, -0.933330]);
```

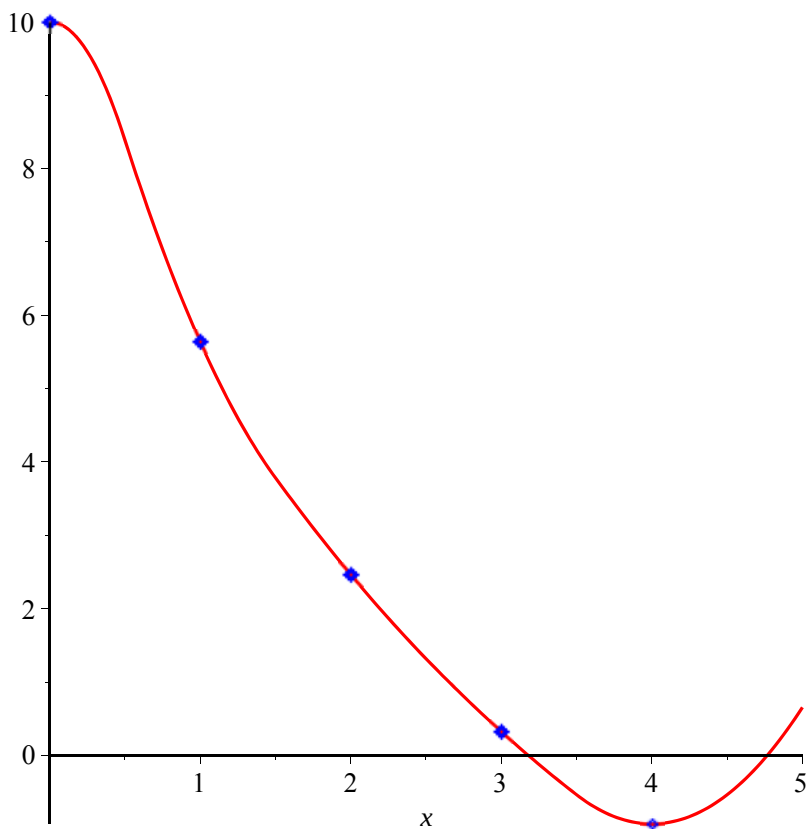
```
Y := [ 10 5.641667 2.466667 0.3250000 -0.933330 ]
```

(34)

```
> f[2] := spline(X, Y, x, quadratic);
```

$$f_2 := \begin{cases} 10. - 6.41111060784314 x^2 & x < 0.500000000000000 \\ 10.2527777843137 - 4.61111078431373 x + 1.79999982352941 (x - 1)^2 & x < 1.500000000000000 \\ 7.40000158823529 - 2.46666729411765 x + 0.344443666666666 (x - 2)^2 & x < 2.500000000000000 \\ 5.89166035294118 - 1.85555345098039 x + 0.266670176470588 (x - 3)^2 & x < 3.500000000000000 \\ -0.933330000000000 + 1.58888327450980 (x - 4)^2 & \text{otherwise} \end{cases}$$

```
> plot([f[2], [X[i], Y[i], i = 1..5]], x = 0..5, style = [line, point], color = [red, blue]);
```



```
> x:=2.3;f[2];
```

```
x := 2.3
1.75766674176471
```

(36)

```
> x:=2.7;f[2];
```

```
x := 2.7
0.905666351176471
```

(37)

```
> x:='x';
```

c) Cubic:

```
x := x
```

(38)

```
> restart;
```

```
> with(plots) :
> X := vector([0, 1, 2, 3, 4]);
```

$$X := \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix} \quad (39)$$

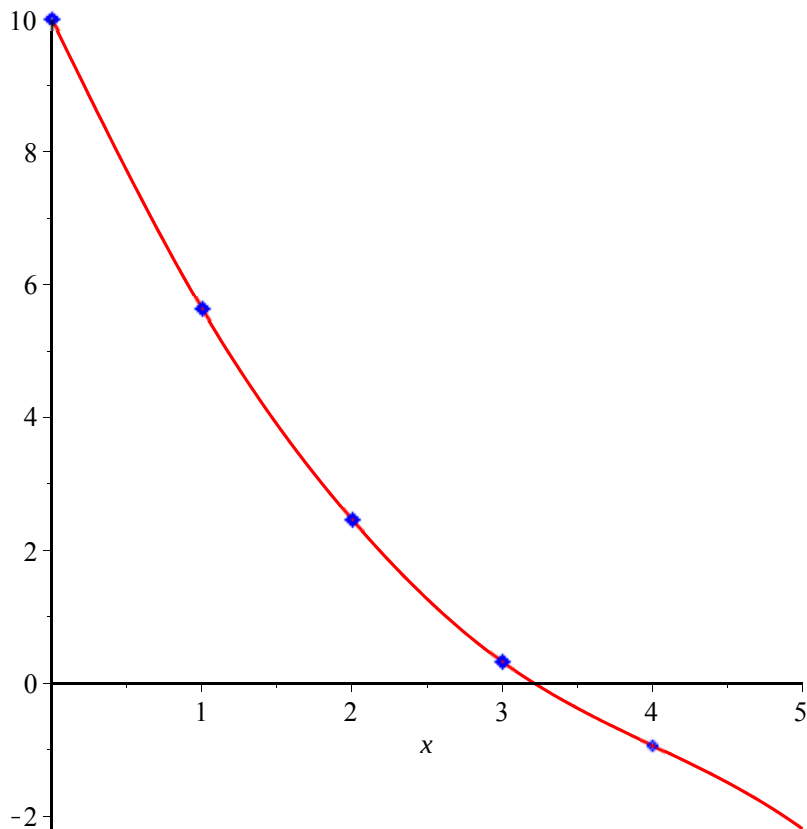
```
> Y := vector([10, 5.641667, 2.466667, 0.3250000, -0.933330]);
```

$$Y := \begin{bmatrix} 10 & 5.641667 & 2.466667 & 0.3250000 & -0.933330 \end{bmatrix} \quad (40)$$

```
> f[3] := spline(X, Y, x, cubic);
```

$$f_3 := \begin{cases} 10. - 4.61726157142857 x + 0.258928571428572 x^3 \\ 9.48214285714286 - 3.84047585714286 x + 0.776785714285715 (x - 1)^2 - 0.111309857142857 (x - \\ 7.708335000000000 - 2.620834000000000 x + 0.442856142857142 (x - 2)^2 + 0.0363108571428574 (x - \\ 5.20356742857143 - 1.62618914285714 x + 0.551788714285715 (x - 3)^2 - 0.183929571428571 (x - \end{cases}$$

```
> plot([f[3], [X[i], Y[i], i = 1..5]], x = 0..5, style = [line, point], color = [red, blue]);
```



```
> x := 2.3; f[3];
```

$$x := 2.3$$

$$1.72125424600000 \quad (42)$$

```
> x := 2.7; f[3];
```

```
|  
|=  
|= > x:='x';  
|=  
|= >
```

$x := 2.7$
0.8615373339999999

(43)